APA
American Planning Association
Technology Division
Creating Great Communities for All

Scan to
Respond to
Session Poll

AI in Planning Hub Discussion

Presentation and Demo | NPC 2026

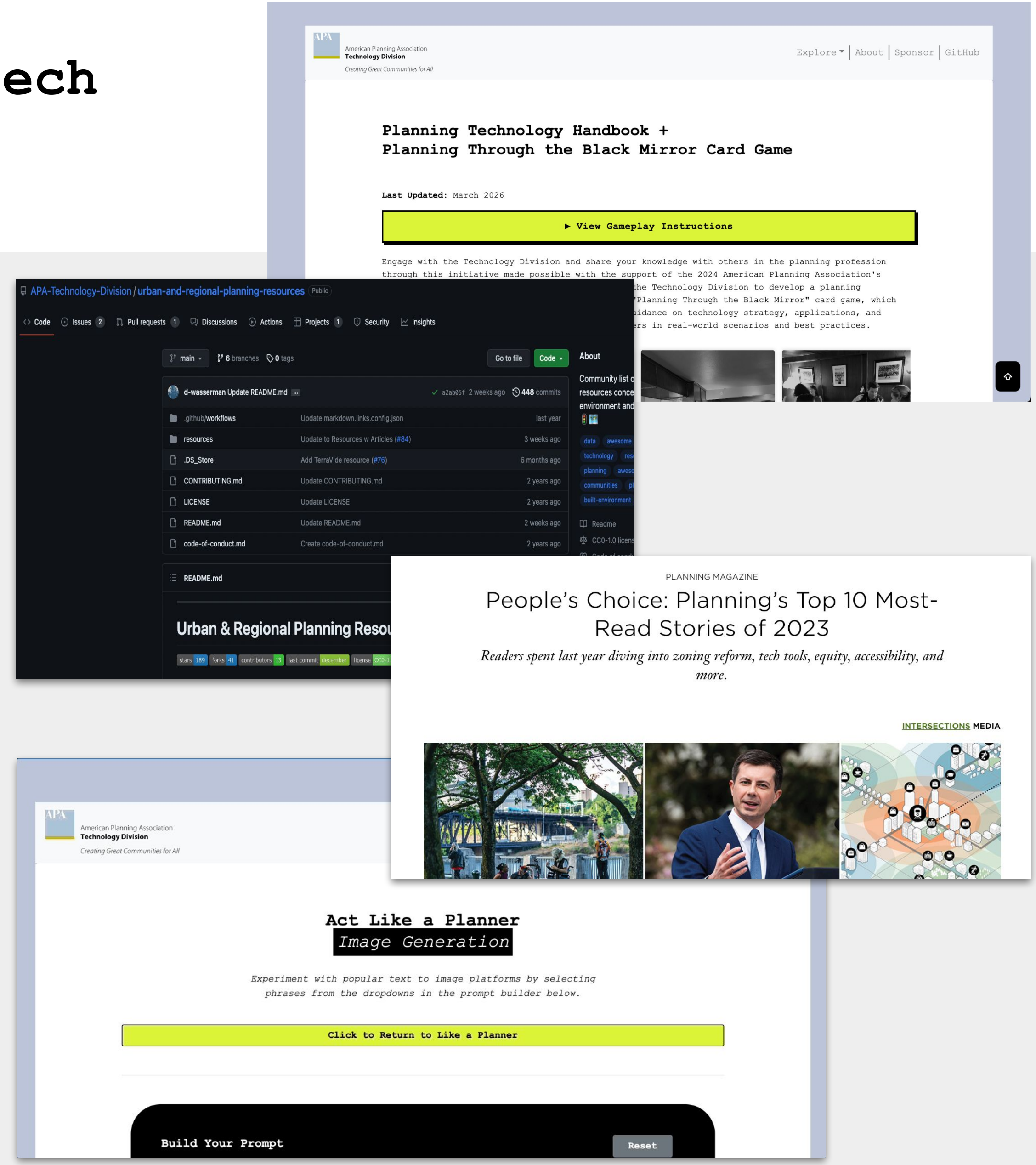
What to Expect

- Introductions
- Updates and survey findings
- Considerations for AI usage
- Quick thought exercise
- Interactive Demo

Scan the Code & Participate the Anonymous Session Poll or use URL:
https://forms.gle/MTcmeEY5wL5uCF7T9

About the Tech Division

- Knowledge exchange
- Member engagement
- NPC programming
- Other in-person events



TECH DIVISION @ NPC26

ATTEND THE ONE WORLD RECEPTION

Monday, 5:30 PM - 8:30 PM

Guardian Building

PLANNING WITH AI

Monday, 8:30 - 9:30 AM

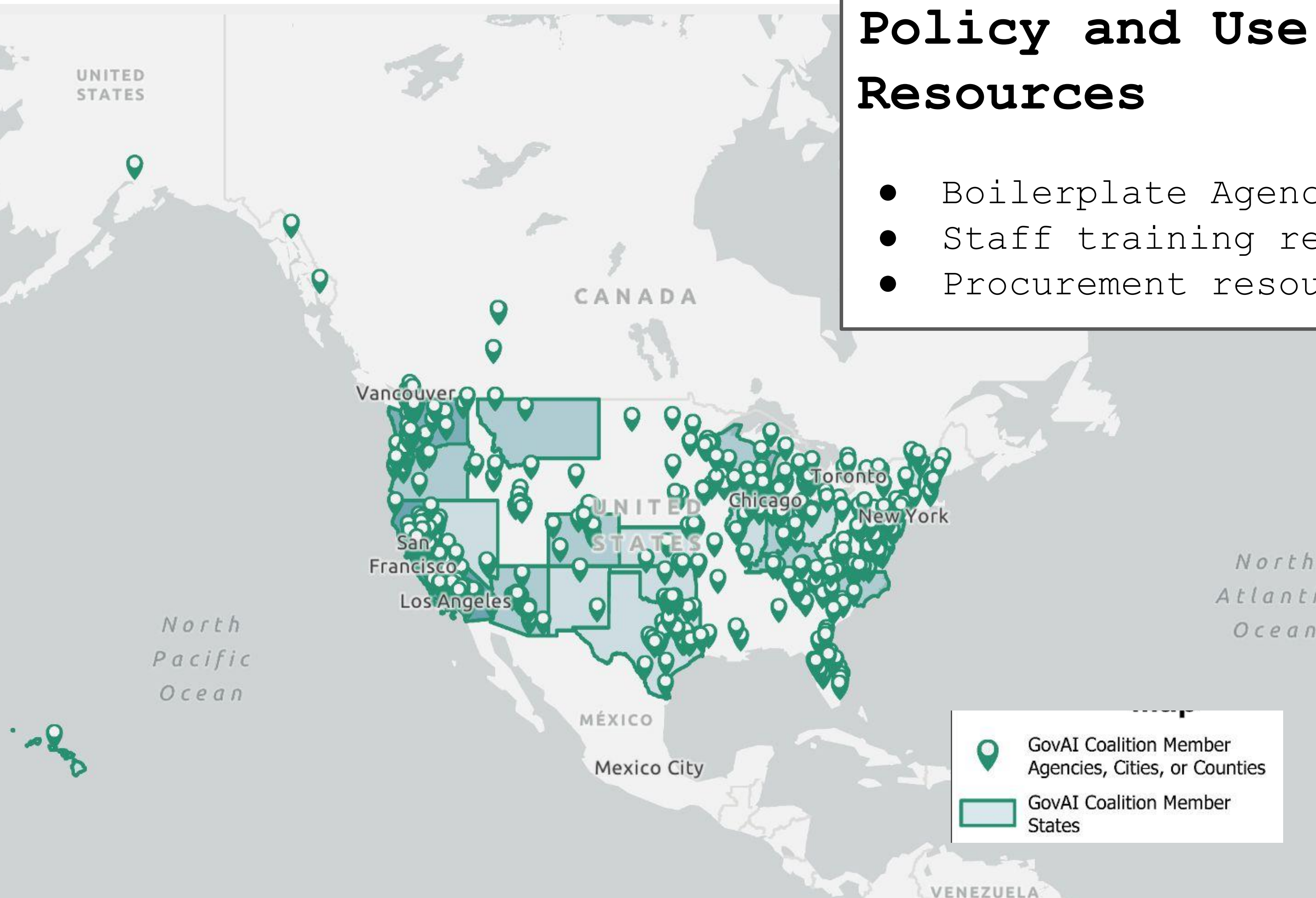
HPCC Rm.310AB

- How are planners using AI?
- Benefits and risks of AI use
- Emerging technology transformation implications

APA Tech

GenAI and Planning

GovAI Coalition



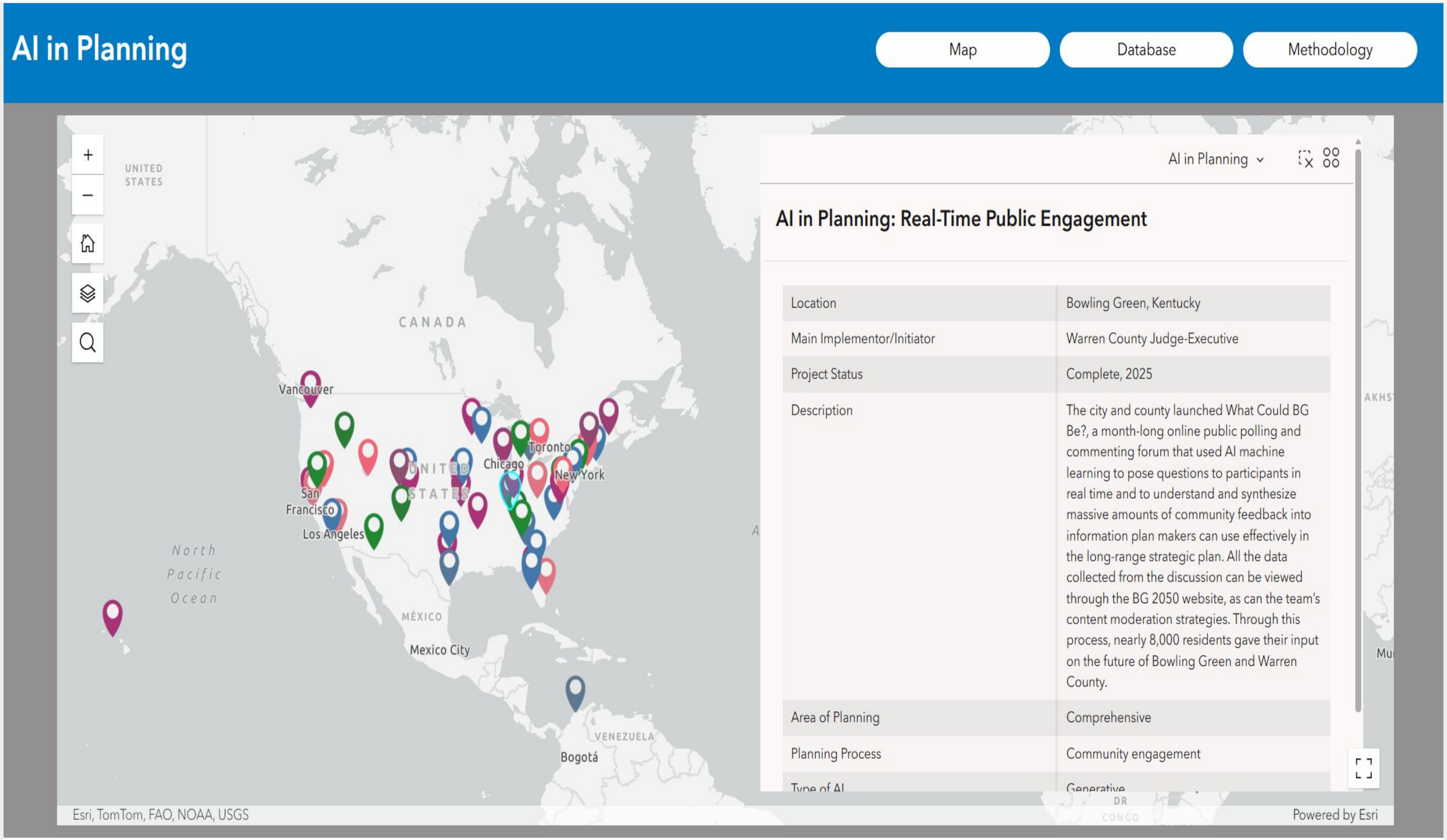
Policy and Use Case Resources

- Boilerplate Agency AI Policies
- Staff training resources
- Procurement resources

AI Darwin Awards

Nominee	Folly (20%)	Arrogance (25%)	Impact (15%)	Lethality (40%)	Final Score
Tesla Full Self-Driving +10	80	90	70	85	92.5
Grok MechaHitler +10	85	95	80	65	91.25
ChatGPT Confidant +10	60	70	75	80	82.25
Tromsø Municipality	80	75	65	35	66
Replit Agent +10	65	70	55	25	65.75
Albania's AI Minister Diella	70	85	60	18	62.75
Omnilert AI Gun Detection	65	75	55	45	60.5
Deloitte Citation Chaos +5 -5	75	70	60	20	60.25
WA Lawyer +5	60	65	45	10	53
Spotify AI Spam Tracks +5	50	60	65	15	51.25
GPT-5 Jailbreak	55	65	50	30	49.25
McDonald's AI Chatbot	75	45	60	15	47.25
MyPillow Lawyers -5	70	60	50	10	44.5

APA Research Agenda Project



APA Research Agenda Project

PlanTech: Updating the Planner's Toolkit

Artificial intelligence (AI) technologies are rapidly becoming embedded in the systems that shape the future of our communities and the everyday work of planners. Across the globe, governments are deploying AI-based tools to meet planning needs. New pilot projects and experiments are popping up constantly, offering promising examples of how AI can expand planning capacity, while also raising important ethical questions, potential risks, and implementation challenges.

These use cases highlight emerging applications of AI in relation to planning in the U.S. and internationally, either in development or currently deployed.

APA continues to monitor emerging use cases of how AI is being used to maintain planning function and plans to provide a comprehensive database in 2026. Stay tuned!

Enforcing Bus Lane Priority

THE PROBLEM: Bus lanes are often contested spaces between cars, bicycles, delivery vehicles, and buses themselves. Police and traffic officers must constantly be on alert for the many standing and parking violations occurring in bus lanes in cities every day. AI-enhanced cameras can help by scanning bus lanes for violations and enhancing traffic control.

USE CASE: Los Angeles offers a nearly yearlong example of how AI cameras can work. The LA Department of Transportation and Metro, LA's public transit system, identified two bus lines that had been "hampered due to parking violators" to pilot their program. The cameras, which use machine learning to identify when cars are parked illegally, were mounted on bus windshields in November 2024. Cameras produce an "evidence package" that includes photos and documentation of vehicle license plates and the date and time of the incident. In February 2025, Metro began issuing tickets. A human is always in the loop; traffic officers review each evidence package before a ticket is issued. By April 2025, Metro had issued nearly 10,000 tickets and expanded the program to two more bus lines. This program further discourages drivers from using bus lanes, improving travel time and experience for transit riders.

Strengthening Wildfire Monitoring

THE PROBLEM: Wildfires are one of the many natural disasters exacerbated by climate change. Many times, wildfires start in remote places where people can't spot them until they spread to disastrous levels. California residents have experienced severe property damage, mental health challenges, and physical health concerns because of devastating fires.

USE CASE: The Orange County Fire Authority and SensoRy AI partnered in 2022 to implement a preventative monitoring system that uses AI to spot new wildfires and alert authorities before they have a chance to spread. In January 2025, the detection system spotted its first fire at 2 a.m. in a remote area of Orange County, where a 911 call about the fire would have been unlikely, and sent an alert to the Orange County Fire Authority. After this successful detection, the system was deployed in multiple locations across Orange County to continue development and monitoring. This detection system has the potential for widespread application across many natural disasters in many locales.

~~How CAN~~ SHOULD GenAI be used to meet these requirements?

(And if so, under what conditions?)

Inventing “a more convenient truth”

Home / Nominees / 2025 Nominees / Tromsø Municipality

Tromsø Municipality - “Closing Schools with Fictional Sources”

Verified

Nominee: Tromsø Municipality and Municipal Director Stig Tore Johnsen for using artificial intelligence to generate research citations for a critical school closure report, creating a policy foundation built entirely on fabricated academic sources.

Reported by: NRK investigation with follow-up reporting by David Gerard and multiple Norwegian outlets - March 28, 2025.

The Innovation

Tromsø Municipality faced the challenging task of justifying the closure of eight schools and several kindergartens—a decision that would affect thousands of families and reshape the city's educational landscape. Rather than conduct thorough research using actual academic sources, the municipal administration discovered the efficiency of artificial intelligence assistance. They confidently deployed AI to help create a comprehensive 120-page report that would serve as the foundation for one of the most significant educational policy decisions in the municipality's recent history. The report needed robust academic backing to convince sceptical residents and politicians that school closures were justified, so naturally, they turned to technology that specialises in producing convincing-sounding content.

The Fabrication Festival

The municipality's spectacular display of confidence in AI-generated research resulted in a report where only seven of 18

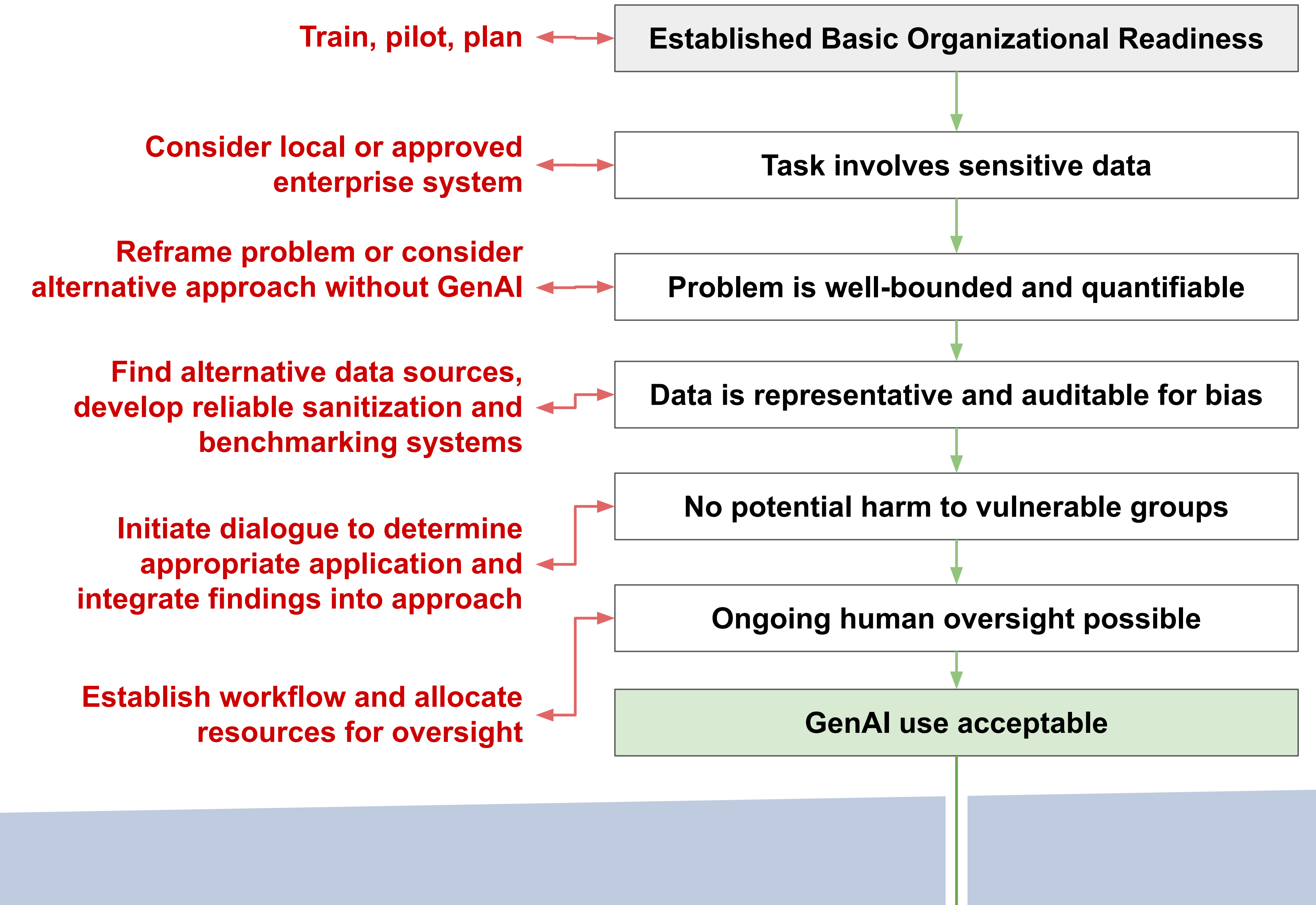
Scenario:

You work for a municipality that is undergoing a major overhaul of its zoning code.

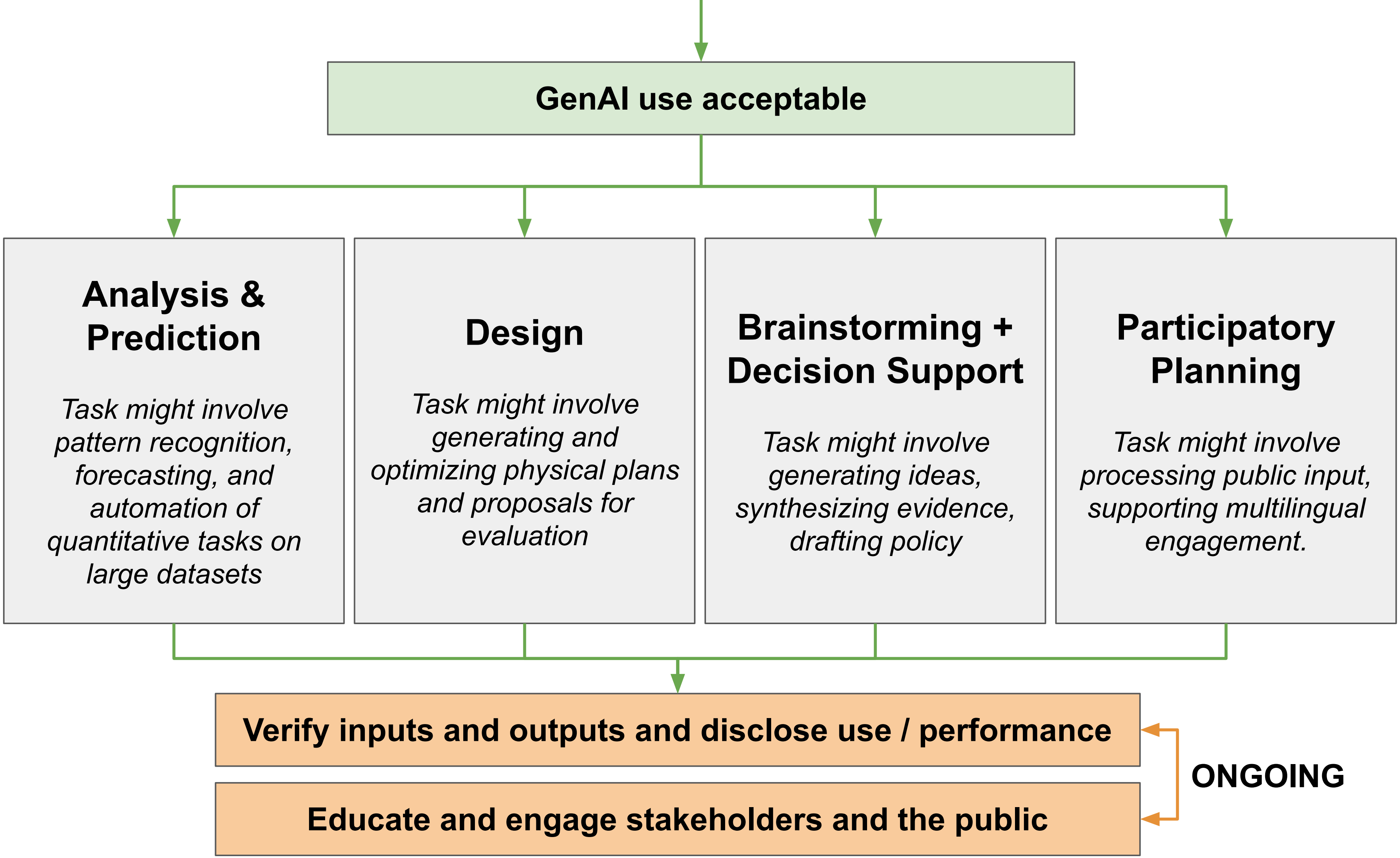
Your department is working on an RFP for a chatbot to answer specific questions from the public about the code update.

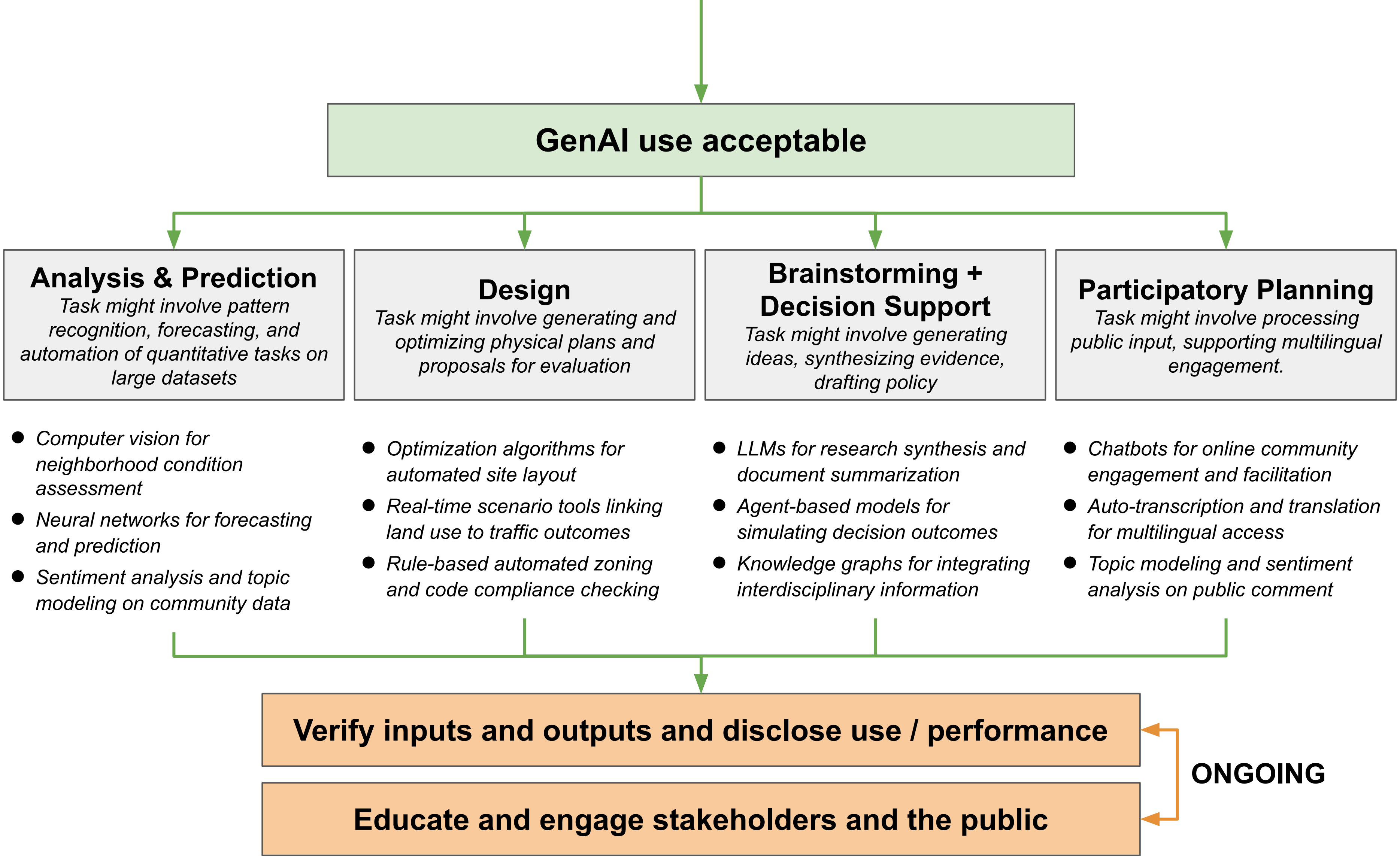
Requirements:

- **Answers questions** about permitted uses, setbacks, height limits, and parking requirements for any zoning district
- **Returns answers in laymans terms**
- **Cites the specific code section** corresponding to each answer
- **Informs users when a question is outside the scope** of the zoning code and direct them to planning staff
- **Reflects the most current version of the code.**
- **Supports follow-up questions** within the same session
- **Is accessible** on the municipal website without login, including on mobile devices
- **Allows users to save** or print a session transcript



Train, pilot, plan
Established Basic Organizational Readiness
Consider local or approved enterprise system
Task involves sensitive data
Reframe problem or consider alternative approach without GenAI
Problem is well-bounded and quantifiable
Find alternative data sources, develop reliable sanitization and benchmarking systems
Data is representative and auditable for bias
Initiate dialogue to determine appropriate application and integrate findings into approach
No potential harm to vulnerable groups
Establish workflow and allocate resources for oversight
Ongoing human oversight possible
GenAI use acceptable


GenAI use acceptable
Analysis & Prediction
Task might involve pattern recognition, forecasting, and automation of quantitative tasks on large datasets
Design
Task might involve generating and optimizing physical plans and proposals for evaluation
Brainstorming + Decision Support
Task might involve generating ideas, synthesizing evidence, drafting policy
Participatory Planning
Task might involve processing public input, supporting multilingual engagement.
Verify inputs and outputs and disclose use / performance
Educate and engage stakeholders and the public
ONGOING


GenAI use acceptable
Analysis & Prediction
Task might involve pattern recognition, forecasting, and automation of quantitative tasks on large datasets
Design
Task might involve generating and optimizing physical plans and proposals for evaluation
Brainstorming + Decision Support
Task might involve generating ideas, synthesizing evidence, drafting policy
Participatory Planning
Task might involve processing public input, supporting multilingual engagement.
Computer vision for neighborhood condition assessment
Neural networks for forecasting and prediction
Sentiment analysis and topic modeling on community data
Optimization algorithms for automated site layout
Real-time scenario tools linking land use to traffic outcomes
Rule-based automated zoning and code compliance checking
LLMs for research synthesis and document summarization
Agent-based models for simulating decision outcomes
Knowledge graphs for integrating interdisciplinary information
Chatbots for online community engagement and facilitation
Auto-transcription and translation for multilingual access
Topic modeling and sentiment analysis on public comment
Verify inputs and outputs and disclose use / performance
Educate and engage stakeholders and the public
ONGOING

American Planning Association Technology Division. (2023). An open letter to the planning community regarding the ethical use of artificial intelligence in planning. https://apa-technology-division.github.io/2023-ai-ethics-letter.html

GovAI Coalition. (2024, March 13). GovAI Coalition templates and resources. City of San José. https://www.sanjoseca.gov/your-government/departments-offices/information-technology/artificial-intelligence-inventory/govai-coalition/templates-resources

Du, J., Ye, X., Jankowski, P., Sanchez, T. W., & Mai, G. (2024). Artificial intelligence enabled participatory planning: a review. International Journal of Urban Sciences, 28(2), 183-210.

Sanchez, T. W., Brenman, M., & Ye, X. (2025). The ethical concerns of artificial intelligence in urban planning. Journal of the American Planning Association, 91(2), 294-307.

Sanchez, T. W., Shumway, H., Gordner, T., & Lim, T. (2023). The prospects of artificial intelligence in urban planning. International journal of urban sciences, 27(2), 179-194.

Reviewing the Requirements…

Requirements:

- Answers questions
- Answers in laymans terms
- Cites the specific code section
- Informs users when a question is outside the scope
- Reflects the most current version of the code.
- Supports follow-up questions
- Is accessible
- Allows users to save

Considerations:

- **Organizational readiness**
- **Sensitive data**
- **Well-bounded problem**
- **Representative and auditable data**
- **Harm to vulnerable groups**
- **Human oversight**
- **Verify and disclose**
- **Educate and engage**

APA Tech

DEMO

Laptop Access:

- **Navigate** to tech.planning.org
- **Click** on the link to the APA Technology Division GitHub Landing Page
- **Select** the "NPC 2026…" folder in the apa-technology-webinar-resources repository

Mobile Access:

Scan to access demo. You can follow along on your phone using this QR code!

> Select the file:
Tech_division_NPC_AI_Hub_Session.ipynb

> Click on the "Open in Colab" button at the top of the file preview

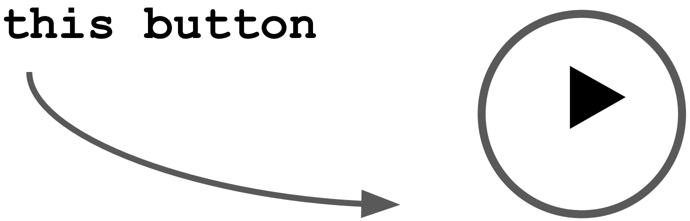
> Click "Connect"

> Scroll down to Part 0

Part 0

Set Up Your Environment

You'll be running code cells by clicking this button



Demo Part 1

Small Dataset Exploration

Scenario:

You work for a municipality that is undergoing a major overhaul of its zoning code.

Your department is working on an RFP for a chatbot to answer specific questions from the public about the code update.

You have launched the chatbot and are now receiving feedback from users about its performance!

Small Dataset (1)

Goals:

1) (Satisfaction) How satisfied are you with this solution (1-5)

2) (Issue Rate) Did you encounter any issues with the response? (0/1)

3) (Diagnosis; solutioning) If yes, please describe the issue you encountered in brief.

📎 **[ATTACH CSV FILE]**

"Based on this data, tell me about 1) user satisfaction over time, 2) the rate at which issues are popping up, 3) the types of issues that are popping up."

Open in Colab

Note: You can copy this text directly from the notebook file

REVIEW OF OUTPUTS

chatbot_public_feedback.csv

CSV

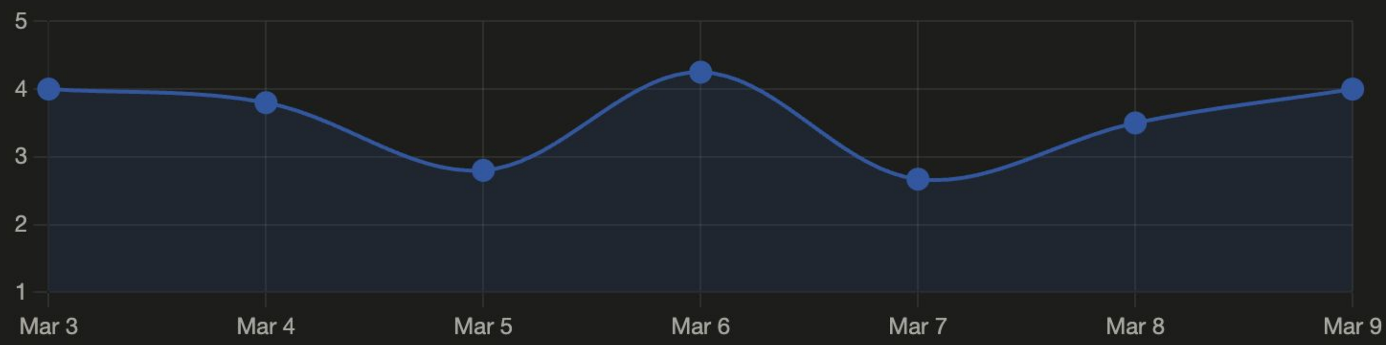
Based on this data tell me about 1) user satisfaction over time, 2) the rate at which issues are popping up, 3) the types of issues that are popping up.

1:48 AM

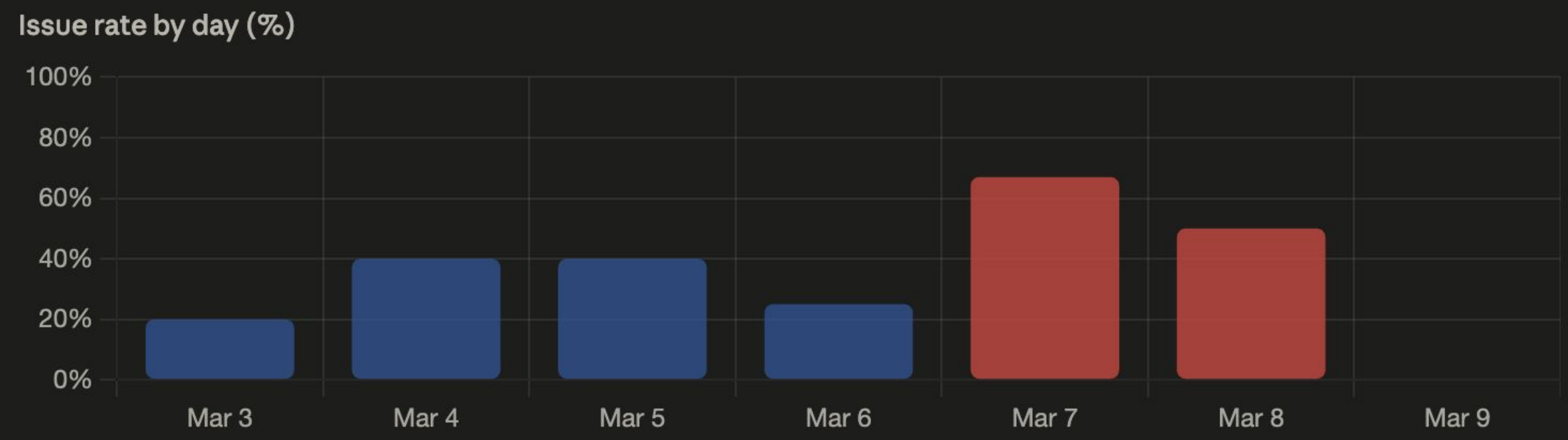
Synthesized survey metrics into multi-panel dashboard visualization >

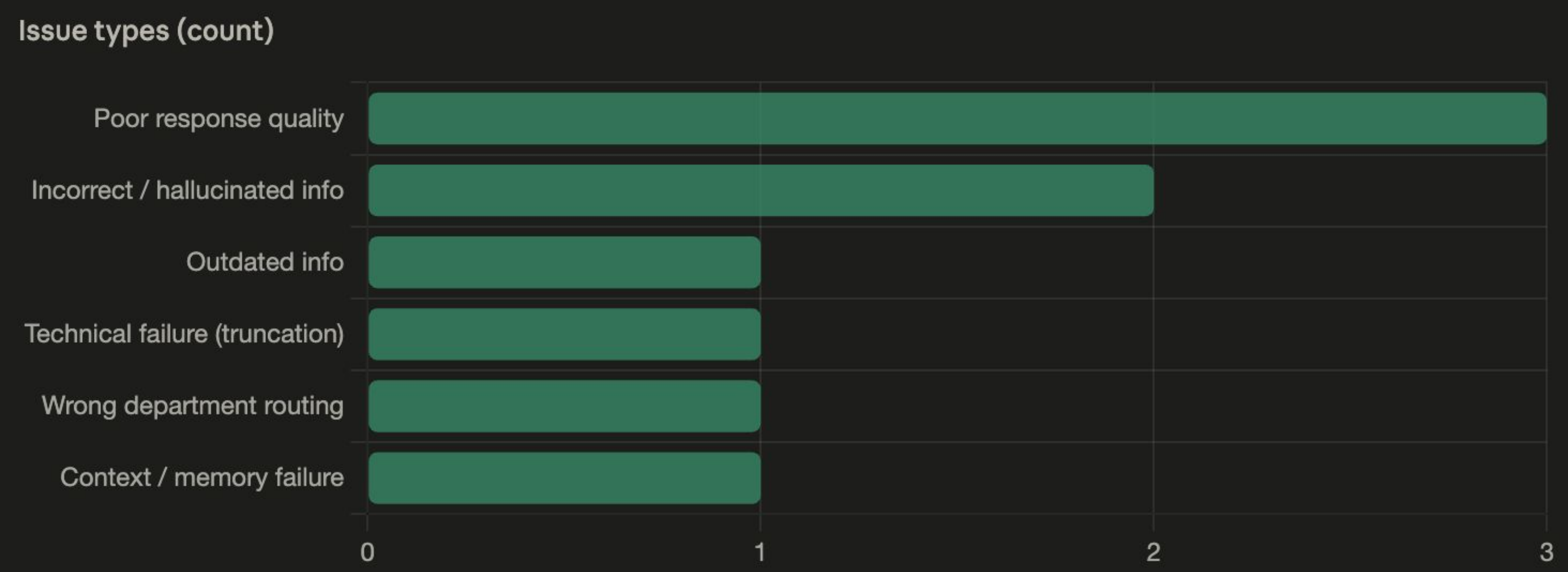
Avg satisfaction	Total responses	Issues reported	Overall issue rate
3.5 / 5	25	10 of 25	40%

Avg satisfaction score by day



REVIEW OF OUTPUTS





REVIEW OF OUTPUTS

3. Types of issues

Ten issues were reported, and they fall into six distinct categories:

- **Poor response quality (3 issues):** This is the most common bucket — one user got a vague, off-topic answer about court archives; another got a response that looped jurisdictions incorrectly (city vs. state); a third got a repetitive, padded non-answer about processing times.
- **Incorrect / hallucinated info (2 issues):** One user received a confidently stated broken URL; another was given a wrong fee amount with no uncertainty expressed. Both are high-trust failures because the chatbot presented wrong facts as certain.
- **Outdated info, technical failure (truncation), wrong department routing, and context/memory failure** each account for 1 issue. The truncation issue is worth flagging — the same broken response reproduced on retry, suggesting a systematic bug rather than a one-off.

REVIEW OF OUTPUTS

how were the types of issues identified

The issue types were identified by me — I read each of the 10 free-text `issue_description` entries in the CSV and manually categorized them based on what the user described. Here's how each one was classified:

Response	Description summary	Category assigned
R003	Chatbot gave a confident URL that was broken/nonexistent	Incorrect / hallucinated info
R006	Chatbot said mail-only submission, but online has been available for a year	Outdated info
R009	Asked about historical court archives, got a generic public records answer	Poor response quality

Run the Remaining Code Cells in Part 1 of the Notebook

Calculated the mean user satisfaction score overall and daily mean user satisfaction score and visualized the result

Calculate and plot the daily issue rate

Parse out and review specific issues from users

Demo Part 2

The Context Window Problem

Demo Part 3

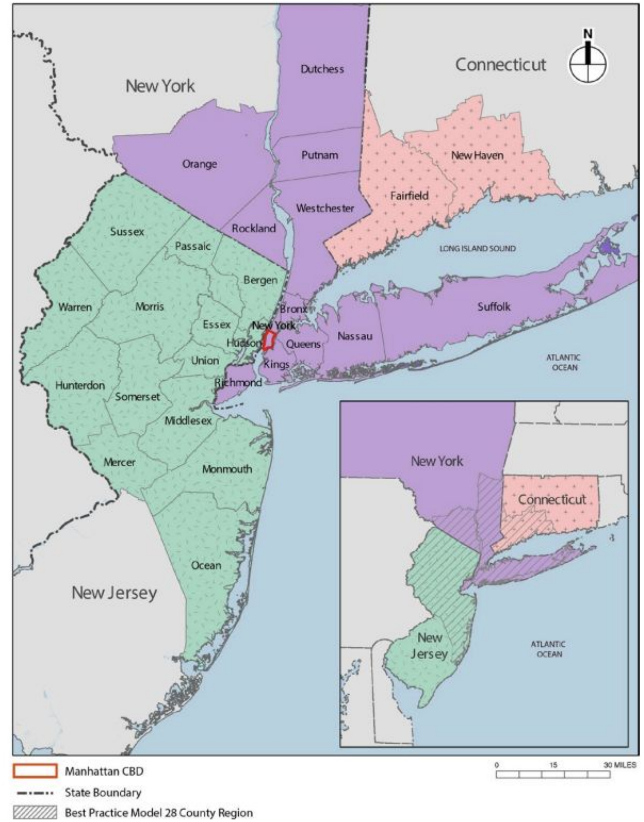
Large Dataset Exploration Using NLP for Reproducible Results

Scenario:

Need to review thousands of public comments for the Manhattan Congestion Relief Zone Tolling Program Environmental Assessment

Conducted to evaluate potential adverse impacts and identify alternatives + mitigation measures.

Figure ES-1. The 28-County Region Study Area



Source: ESRI, NYC Open Data, NYMTC 2020 TransCAD Highway Network

Source: [MTA]

Large Dataset (2)

Goals:

1. **(Overall sentiment toward the project)** What is the general sentiment toward the project across comments in the dataset
2. **(General concerns)** What themes or concerns are associated with positive, negative, or neutral responses?
3. **(Fairness concerns)** Do commenters believe the proposed mitigation measures are sufficient for low-income drivers?

Large Dataset (2)

[ATTACH CSV FILE]

"Based on this data tell me about

1) what designs are considered most feasible,
2) what designs are most favored,
3) how do people generally feel about the project,
4) what are some shared concerns among community members about the project/location?"

Open in Colab

Note: You can copy this text directly from the notebook file

REVIEW OF OUTPUTS

anonymized_survey_results - anonymized_survey_results.csv

CSV

Based on this data tell me about 1) what designs are considered most feasible, 2) what designs are most favored, 3) how do people generally feel about the project, 4) what are some shared concerns among community members about the project/location?

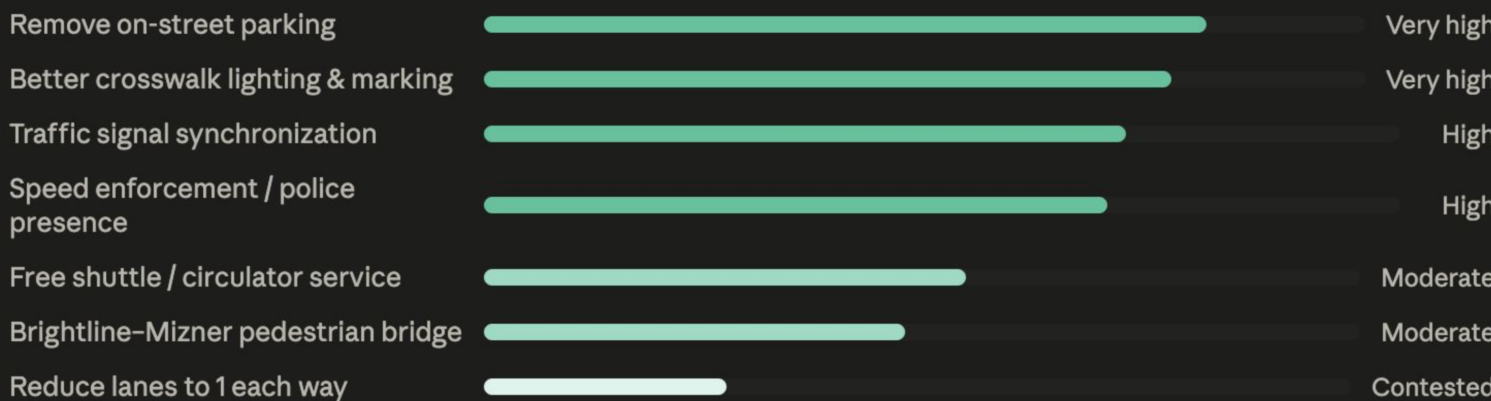
REVIEW OF OUTPUTS

Synthesized survey findings across feasibility, preference, sentiment, and community concerns >

Now I have a thorough read of all 1,639 responses. Here's my analysis:

Analysis of 1,639 survey responses

1 — MOST FEASIBLE DESIGNS



Bars reflect relative frequency of support across open-ended responses — not formal voting.

2 — MOST FAVORED DESIGNS

Atlantic Ave / Las Olas model — the clear frontrunner

By far the most-cited design inspiration. Respondents consistently point to Delray Beach's Atlantic Ave and Fort Lauderdale's Las Olas as the vision: one or two narrowed lanes, wider sidewalks, heavy outdoor dining, slower traffic, and a walkable "boulevard" feel. Many feel Springfield has fallen behind and see this as the obvious path.

REVIEW OF OUTPUTS

3 — GENERAL SENTIMENT TOWARD THE PROJECT

~45%	~35%	~20%
Hopeful — want improvements, engaged with ideas	Frustrated — feel it's too late, city caused the problem	Skeptical or opposed — don't touch lanes / it's fine as is

Hopeful but conditional

Many supporters want improvements but worry this survey will go nowhere, like previous ones. There's deep skepticism about follow-through and a sense that city government has historically prioritized developer interests over residents.

> *"I have been involved in studies like this 3 or 4 times before. We have made a few improvements over the years, however, there is no perfect solution."*

Strong opposition to lane reduction

A vocal minority is firmly opposed to reducing vehicle lanes, citing evacuation routes, existing congestion, and island residents' need for beach access. Many warn that narrowing Evergreen Terrace will push traffic onto already-overwhelmed parallel streets like Camino and Spanish River.

> *"DO NOT reduce or eliminate driving lanes! The current bridge area congestion will only get worse due to poor planning years ago."*

REVIEW OF OUTPUTS

for each of the items in the analysis explain how the metrics were derived

Acknowledged qualitative synthesis methods and transparency limitations >

This is an important question, and I want to be fully transparent: **the metrics in the visualization were qualitative estimates, not computed counts.** Here's exactly what I did and where that creates limitations:

The data structure

The CSV has four columns. Two contain open-ended text: a free-form "other priorities" field and a longer narrative response. There were no structured multiple-choice results visible in the data — only free-text. That means there was no column to tally like "X% selected bike lanes." Everything I showed was synthesized from reading the narrative responses.

Feasibility bars (section 1)

Labels like "Very high," "High," "Moderate," and "Contested" were my judgment calls based on how frequently a design appeared in the text and how much opposition it attracted. I did not run a keyword count. "Remove on-street parking" being rated highest reflects that I encountered it very frequently *and* it attracted less pushback than, say, lane reduction. But I didn't count instances — I estimated after reading. The bar widths (82%, 78%, etc.) were illustrative, not computed.

Run the Remaining Code Cells in Part 2 of the Notebook

Sentiment analysis using VADER; visualization using Plotly

Topic modeling using LDA with visualization

Enhanced word frequency analysis supported by LLM with visualization

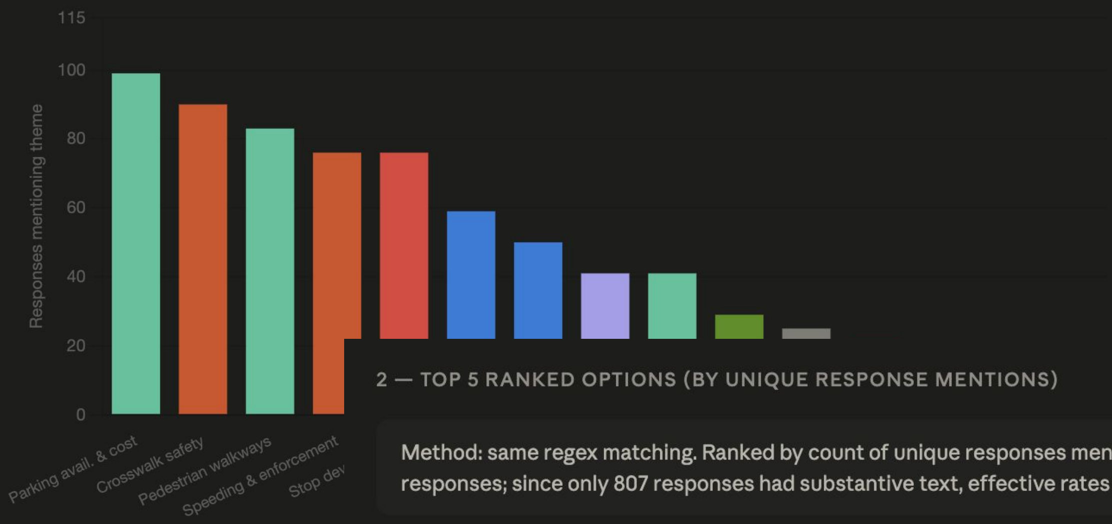
Assessed data limitations and architected comprehensive multi-panel visualization dashboard >

Good — now I have real computed numbers. Let me build the full visualization.

n = 1,639 total responses · 807 with substantive text · methodology notes below each section

1 — Q1 WRITE-IN THEMES: MENTION COUNTS

Method: regex pattern matching on the Q1 "Other (please specify)" + Q2 open-ended fields combined. Each response counted once per theme regardless of how many times the theme appears in that response.



2 — TOP 5 RANKED OPTIONS (BY UNIQUE RESPONSE MENTIONS)

Method: same regex matching. Ranked by count of unique responses mentioning each theme. Percentages are of all 1,639 responses; since only 807 responses had substantive text, effective rates among writers are roughly double.

1	Parking availability & cost	99 (6.0%)
2	Pedestrian crosswalk safety	90 (5.5%)
3	Pedestrian walkways / sidewalks	83 (5.1%)
4	Speeding & enforcement	76 (4.6%)
4	Stop / limit new development	76 (4.6%)

Items 4 and 5 are tied at 76 mentions each. Atlantic Ave / Delray model and bridge construction delays follow closely at 41 mentions (2.5%) each.

3 — SENTIMENT ANALYSIS: Q2 OPEN-ENDED RESPONSES

Method: VADER (Valence Aware Dictionary and sEntiment Reasoner) applied to 750 non-trivial Q2 responses. Compound score ⩾ 0.05 = Positive, ⩽ −0.05 = Negative, otherwise Neutral. VADER is lexicon-based and well-suited to short, opinionated text. Mean compound: 0.094 (slight lean positive). Median: 0.000 (balanced).

46.8%	20.3%	32.9%
Positive	Neutral	Negative
351 of 750	152 of 750	247 of 750

Intensity breakdown (compound score ranges):

Caveat: VADER scores surface-level tone, not nuanced intent. A response saying "it's great but the traffic is terrible" may score positive due to "great" while embedding serious criticism. Treat this as directional, not definitive.

4 — NER: ISSUES CLASSIFIED BY CATEGORY

Method: keyword-pattern taxonomy applied across all 1,639 concatenated responses. Issues were hand-defined using 7 categories, each with 4–9 named issue types and multi-term regex patterns. One count per response per issue. Responses with no text are included in the denominator, so all percentages are conservative.

Safety

Pedestrian crosswalk danger	83 (5.1%)
Speeding vehicles	54 (3.3%)
Cyclist safety	1 (0.1%)
Noise / modified vehicles	22 (1.3%)
Valet / double parking hazards	17 (1.0%)
Intersection visibility	10 (0.6%)

Development & Planning

Overdevelopment / high-rises	72 (4.4%)
Retail / business mix	20 (1.2%)
Loss of small-town character	17 (1.0%)
Zoning / developer influence	6 (0.4%)
Infrastructure capacity / growth	2 (0.1%)

Mobility & Access

Quality of Life

Noise from vehicles / events	24 (1.5%)
Events / entertainment	8 (0.5%)
Pet-friendly amenities	6 (0.4%)
Resident quality of life	4 (0.2%)
Affordability / pricing	4 (0.2%)

Note on "cyclist safety" low count (1): respondents describe bike danger in general terms ("death trap," "not safe," "dangerous to bike") captured under "bike lane access" rather than with explicit cyclist-safety phrasing — not an absence of concern.

Here's what changed from the previous analysis and why it matters:

What's different now — and why

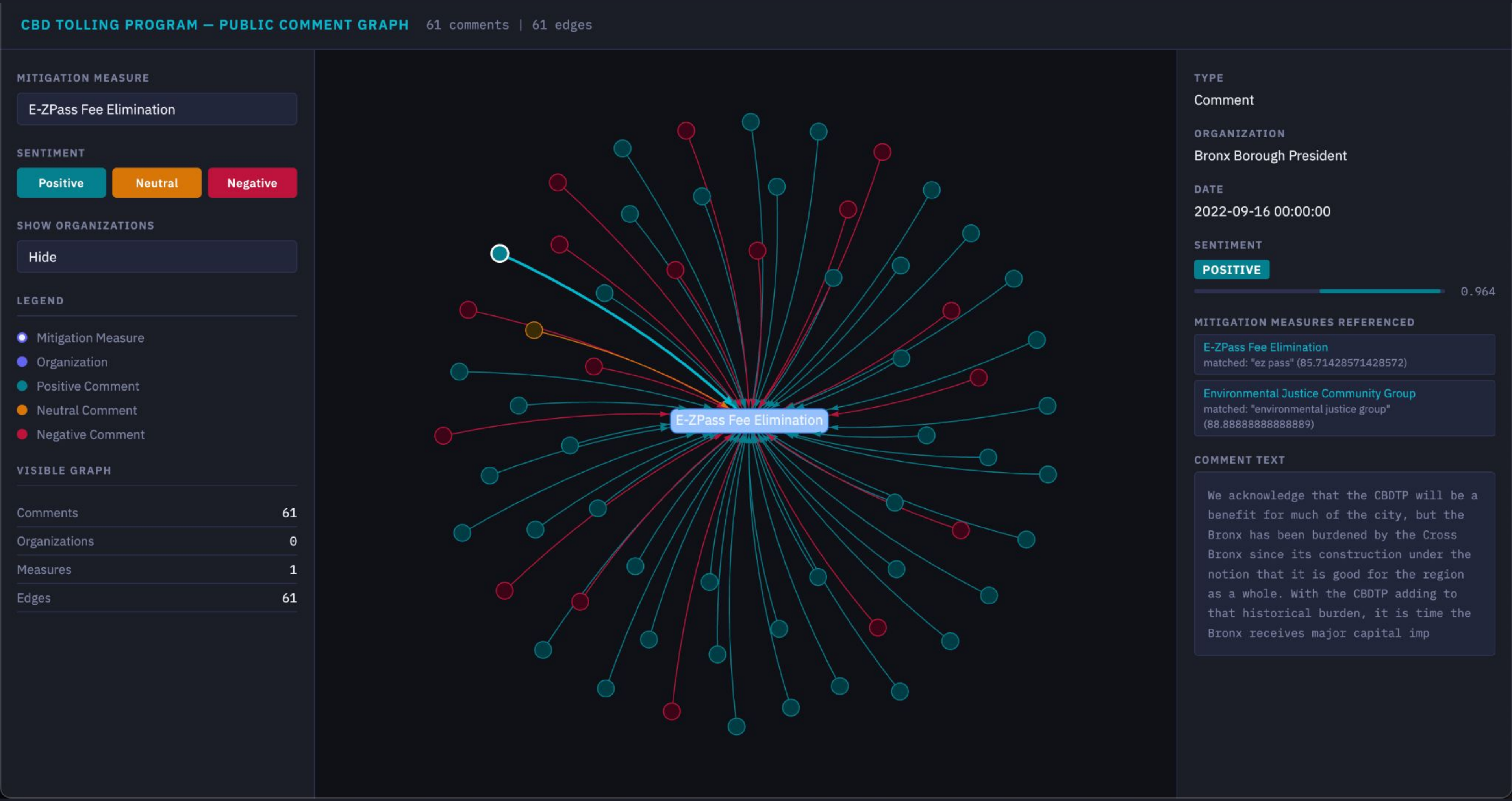
Every number above is computed directly from the text. The methodology for each section:

Q1 theme counts — regex pattern matching across both the "Other (please specify)" write-in field and the Q2 open-ended field, combined. Each response is counted once per theme, so one person writing "we need bike lanes and wider sidewalks and to stop building condos" contributes one count to three different themes. The denominator is always 1,639 (total responses), making all rates conservative since only 807 had substantive text.

Top 5 ranking — a genuine tie emerged at position 4 between "Speeding & enforcement" and "Stop/limit new development" (76 each). That tie was invisible in the previous qualitative estimate.

Sentiment — VADER applied to 750 non-trivial Q2 responses. The 46.8% positive finding is real but needs context: VADER scores the *tone* of individual words and phrases, not the underlying civic intent. Many "positive" responses are critical in substance but use polite or hopeful framing. The bimodal distribution (large Very Positive and Very Negative clusters, with a smaller middle) reflects that people felt strongly in both directions.

NER / issue taxonomy — 7 categories with 37 named issue types, each backed by multi-term regex patterns. The key correction from before: "Cyclist safety" scored just 1 match not because biking is unimportant, but because respondents use descriptive language ("death



TALKING TO YOUR DATA...

Poll Results

Scan to
Respond to
Session Poll

Q&A

More Division Resource for AI and Planning

- AI Ethics Blog Post and Open Letter
- Planning Through the Black Mirror Game
- Archived and Upcoming Webinars and Workshops
- Act Like a Planner